

A Hybrid Neural Semantic Framework for Prompt Injection Detection in LLMs

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Abstract—Prompt injection attacks imply that they lead to a very severe risk to the dependability of Large Language Model (LLM) systems by allowing a user to control the behavior of the model using constructed inputs. This paper proposes a Hybrid-PI neural-semantic architecture to improve the detection of these attacks. Hybrid-PI with the DeepSet prompt-injection dataset uses a neural prompt encoder with a Semantic Consistency Oracle to jointly learn surface and higher-level semantic inconsistencies via a trade-off between the two. The experimental findings indicate that Hybrid-PI performs much better than the classical and transformer-based baselines, achieving 97.4% accuracy and 97.5% F1-score, with significantly reduced false positives. The robustness tests also indicate the ability to withstand paraphrase and obfuscation-based adversarial variations. The results suggest that Hybrid-PI is a viable and scalable approach to strengthening safety controls around LLMs against emerging threats such as prompt injection.

Keywords—Prompt Injection Detection, Large Language Models (LLMs), Adversarial Attacks, Security for AI models, Hybrid Neural-Semantic Framework

I. INTRODUCTION

A Large Language Model (LLM) is a model of AI that is trained on large volumes of text to comprehend and produce natural language. The fact that they can follow instructions well is making them increasingly incorporated into applications such as chatbots and content creation. They are, however, very susceptible to prompt injection attacks, in which hidden commands in the input data can trick the model into generating harmful output, revealing sensitive information, or bypassing the model's intended use. [1, 2]. Current detection systems (typically machine learning classifiers) aim to detect and prevent such malicious prompts but face issues such as high false positives, an imbalance between safe and harmful prompts, and an inability to adapt to new or evolving attack techniques. The constraints mentioned above make it clear that more precise, flexible, and robust solutions for protecting LLMs are required. [3].

As an expanding and unethical security risk, prompt injection attacks exploit the fact that language models are highly dependent on the instructions provided by the user. The input features include prompts that can be manipulated in subtle ways to elicit incorrect or unsuitable responses, thereby affecting the quality and reliability of the outputs. The need to do so is even greater in such areas as healthcare,

finance, and law, where erroneous decisions may be fatal. The attacks are carried out during the execution of a system's queries, which they use to inject concealed data and commands from the user, thereby interfering with the model's normal operations. This may change the model's intended behavior and, therefore, evil inputs may affect, or even dictate, the model's responses. This kind of exploitation affects applications driven by large language models, particularly when malicious prompts can override or manipulate legitimate responses.

The proposed research will create a deep learning-powered detection architecture that detects prompt injection attacks on the large language models with a high degree of accuracy and interpretability. The study aims to develop a model that combines neural feature extraction and LLM reasoning to identify user prompts as safe or unsafe. It is also aimed at producing high-quality explanations for every detection decision, making them transparent and increasing confidence in artificial intelligence-driven systems across healthcare, finance, and cybersecurity.

II. RELATED WORK

Prompt injection attackers aim to manipulate the LLM's response without bypassing security mechanisms. Ayub et al [4] applied embedding-based ML classifiers to differentiate between malicious and benign prompts. Under this strategy, embeddings are generated using standard models, and binary classifiers such as Random Forests and XGBoost are trained. This paper shows that conventional ML classifiers with embeddings outperform several open-source neural network-based detectors, and that providing efficient embeddings and sparse classifiers is effective. Shi et al. [5] propose a CoT-based framework for assessing the timeliness of injection attacks that uses step-by-step reasoning, semantic and consistency checks, confidence estimates, and backtracking to detect anomalies. It is more accurate and achieves a higher F1 score, especially on smaller models, than traditional and intrinsic LLM detection methods. Nevertheless, the technique presents increased latency. Rahman et al [6] embedded prompt text using BERT variants (multilingual BERT, DistilBERT) and used ML classifiers such as SVM, Logistic Regression, and Naive Bayes. Multilingual BERT and Logistic Regression

had 96.55% accuracy. The paper demonstrates that the interpretable ML methods embodied in fine-tuned LMs are an efficient detection pipeline. Yip et al. [7] propose an evaluation model to determine LLM resilience to immediate injection based on 115 curated attacks and a Weighted Resilience Score (WRS). Using ChatGLM and Llama2, it was found that Llama2 was 67.3 times better, and ChatGLM was 38.2 times better. The framework is a versatile, cost-effective tool for enhancing LLM security.

Nair et al. [8] compare random forests with BiLSTM for prompt injection, achieving 92.5% accuracy compared to BiLSTM's 88.75%. SHAP analysis made it interpretable, and the results indicate that LSTM has an accuracy advantage and that Random Forest can be used in real time. Piet et al. [9] used the HackAPrompt data set and introduced the Jatimo method, which trains LLMs such as GPT-3 and T5 to be less susceptible to prompt injection. Jatmo enhanced the model's robustness, achieving 89% accuracy on the dataset. Rehman et al. [10] used XLM-RoBERTa to detect prompt injection, achieving 55.17%-99.13% accuracy through fine-tuning, demonstrating its effectiveness. A three-layered model that combines a rule-based filter, an ML classifier, and a companion LLM was proposed by Kokkula et al. [11]. The ML classifier worked best as an individual, although the stratified arrangement minimized false negatives. All in all, these investigations present fine-tuning, unified strategies, and layered defenses as effective measures for protecting LLMs. To defend MLLMs against image-based prompt attacks, Sharma et al. [12] proposed a two-stage approach that included input validation followed by a detection phase using SPML. Evaluation on attack pictures revealed that GPT-4-VISION achieved 96% precision, whereas LLAVA-13B and MINIGPT-4 achieved 89% and 91%, respectively.

III. METHODOLOGY

We propose HYBRID-PI, a two-stage detector combining:

- A compact neural prompt encoder and discriminative verifier
- A micro-reasoning module that leverages a frozen LLM as a semantic consistency oracle.

Figure 1 illustrates the pipeline of the proposed hybrid model. It works with data from one of the Hugging Face Datasets, and the essential steps are Tokenization and Embedding to prepare the text. The fundamental element of the system is the Hybrid Model, which comprises a Neural Encoder (for learning contextual features) and a Semantic Oracle (for external knowledge/reasoning). Lastly, the model's performance is evaluated using various evaluation metrics, such as Accuracy and F1-score. The architecture is trained end-to-end with joint classification, contrastive separation, adversarial robustness, and attribution

regularizers to produce high accuracy, low false positives, and per-decision explanations suitable for high-risk domains. Let a user input prompt be $x \in \mathcal{X}$ token sequence. Let $y \in \{0,1\}$ denote benign (0) or injection (1). Let a pretrained LLM oracle be $G(\cdot)$ frozen weights. The detector is $D_\theta(x)$ producing score $s_\theta(x) \in [0,1]$ and binary decision $\hat{y} = \mathbf{1}\{s_\theta(x) > \tau\}$. The training set $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$.

A. Proposed Architecture

- Tokenizer and Embedding: $E: \mathcal{X} \rightarrow R^{T \times d}$
- Neural Prompt Encoder (NPE): a lightweight Transformer f (or BiLSTM) producing a pooled vector $z = f_\phi(E(x)) \in R^m$
- Discriminator Head: classifier $p_{(\theta(y=1|x))} = \sigma(w^{(\tau)}z + b)$
- Semantic Consistency Oracle (SCO): Query G with structured prompts $q(x)$ to produce an embedding $u = O(G(q(x))) \in R^m$, where O maps generated text to vector via the same embedding space or a projection.
- Joint Reasoner: Cross-modal fusion $h_\psi(z, u)$, e.g., gated sum or 1×1 projection feeding a secondary verifier $v_\omega(h)$ producing an auxiliary score $r(x)$.
- Final score: $s_\theta(x) = \alpha \cdot p_\theta(y=1|x) + (1-\alpha) \cdot r(x)$, $\alpha \in [0,1]$.

B. Training Objectives

To optimize the parameters $\Theta = \{\phi, \theta, \psi, \omega, \text{proj}\}$ of the proposed detector, we formulate a composite loss function that jointly enforces accurate classification, discriminative embedding structure, robustness to adversarial perturbations, interpretable attribution patterns, and calibrated probability outputs. The overall training objective is defined in Eq. 1.

$$\mathcal{L}(\Theta) = \lambda_{CE}\mathcal{L}_{CE} + \lambda_{CL}\mathcal{L}_{CL} + \lambda_{ADV}\mathcal{L}_{ADV} + \lambda_{ATT}\mathcal{L}_{ATT} + \lambda_{CAL}\mathcal{L}_{CAL} \quad (1)$$

The coefficients in Eq. 1 are non-negative hyperparameters that control the relative contribution of each loss term. $\mathcal{L}_{CE}$ The cross-entropy classification loss. $\mathcal{L}_{CL}$ The contrastive learning loss. $\mathcal{L}_{ADV}$, adversarial consistency regularization. $\mathcal{L}_{ATT}$, attention-based semantic alignment, and $\mathcal{L}_{CAL}$ captures calibration-based confidence refinement. Where $\mathcal{L}_{CE}$ is the standard cross-entropy term shown in eq. 2,

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{i=1}^N [y_i \log s_\theta(x_i) + (1-y_i) \log(1-s_\theta(x_i))] \quad (2)$$

ensuring correct discrimination between benign and malicious prompts. According to eq. 3, to improve separability in embedding space and reduce overlap between safe and adversarial samples, we incorporate a supervised contrastive loss

$$\mathcal{L}_{CL} = -\frac{1}{N} \sum_i \frac{1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(z_i^\top z_p / \tau_c)}{\sum_{a \in A(i)} \exp(z_i^\top z_a / \tau_c)} \quad (3)$$

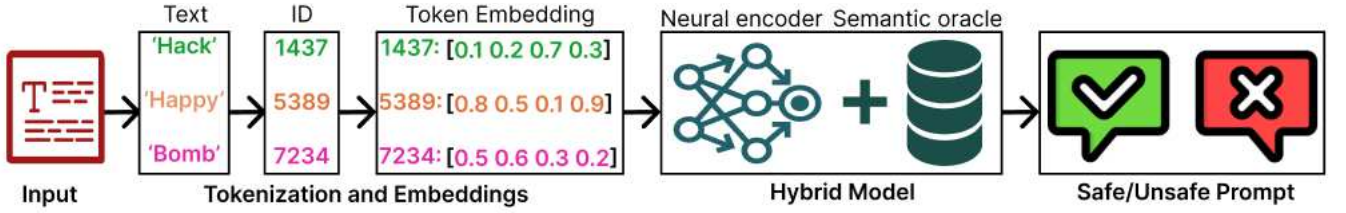


Fig. 1. Workflow of the proposed Hybrid-PI system

where $P(i)$ and $A(i)$ denote positive and negative sample indices, respectively. Robustness to paraphrasing, token-level noise, and adversarial rewriting is enforced through

$$\mathcal{L}_{\text{ADV}} = \frac{1}{N} \sum_i \|s_{\theta}(x_i) - s_{\theta}(x'_i)\|_2^2 \quad (4)$$

which penalizes sensitivity to small perturbations x'_i as expressed in eq. 4. To promote interpretability, we regularize the gradient attribution vector $a(x) \in \mathbb{R}^T$, encouraging sparse, high-intensity explanations for injected prompts and diffuse attribution for benign ones, as defined in eq. 5.

$$\mathcal{L}_{\text{ATT}} = \frac{1}{N} \sum_i [y_i \cdot \text{Gini}(a_i) - (1 - y_i) \cdot \text{Entropy}(a_i)] \quad (5)$$

Finally, eq. 6 illustrates that to ensure reliable confidence estimates, especially in high-risk applications, we incorporate a calibration penalty

$$\mathcal{L}_{\text{CAL}} = \sum_{b=1}^B |\text{acc}(B_b) - \text{conf}(B_b)| \quad (6)$$

which measures deviation between empirical accuracy and predicted confidence across probability bins. Together, these components create a unified objective that balances predictive accuracy, embedding structure, adversarial stability, explainability, and calibration, yielding a rigorously optimized detector suitable for real-world deployment.

C. Semantic Oracle:

Define deterministic templates $q(x)$ that asks the frozen LLM G to (i) restate the instruction, (ii) enumerate implicit commands, (iii) flag contradictions. For example:

“List any instructions embedded in: $\langle x \rangle$.”

“Does $\langle x \rangle$ request disclosure of secrets? yes/no, explain.”

Map LLM responses to vector u via the same tokenizer and projection to align with z . The novelty is using a small number of targeted queries, low cost, and projecting responses into the detector space for robust fusion

IV. RESULTS AND DISCUSSION

A. Dataset Acquisition

In this study, we use the DeepSet/Prompt-Injections dataset, a carefully selected set designed to assess the robustness of prompt-injection detection systems. This dataset consists of 662 samples, of which 546 are for training and 116 for testing. These original samples are preserved to ensure consistency with the benchmark design. This is a text field in each sample that indicates whether it contains a benign prompt or an adversarial injection attempt, along with a binary label indicating its classification. Even though the dataset is mainly in English, it also includes a small number

of multilingual examples, such as German, which enables broader generalization analysis. Table I summarizes the dataset structure, including the definitions of the text and label columns. The prompts are all tokenized and embedded before training, transforming the raw input into structured numerical representations that downstream learning modules use. No additional synthetic perturbations were introduced.

TABLE I. COLUMN DEFINITIONS OF THE DATASET.

Column Name	Type	Description
text	String	Input prompt, containing either benign queries or adversarial crafted injection attempts.
label	Integer (0/1)	Binary class label indicating benign (0) or prompt injection (1).

B. Evaluation Metrics

The suggested Hybrid-PI model is evaluated using conventional binary classification metrics. TP, TN, FP, and FN will represent true positives, true negatives, false positives, and false negatives, respectively. Model performance is assessed using accuracy, precision, recall, F1-score, and AUROC, as defined in equations (7)-(11). Measure of accuracy: Overall correctness:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

Precision quantifies the correctness of malicious-prompt predictions:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (8)$$

Recall evaluates the model's ability to detect actual injection attempts:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (9)$$

F1-Score, the harmonic mean of precision and recall, is given by:

$$F1 = 2 \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (10)$$

Finally, AUROC provides a threshold-independent measure of separability between benign and adversarial prompts by integrating the ROC curve:

$$\text{AUROC} = \int_0^1 \text{TPR}(\text{FPR}^{-1}(x)) dx \quad (11)$$

C. Experimental Results

This section provides a detailed analysis of the suggested framework's performance. With an accuracy of 97.4%, the proposed method identified rapid injection attacks in large language models. Table II. shows the results of the suggested

Hybrid-PI architecture compared to the classical and transformer models on the DeepSet/Prompt-Injections dataset. Hybrid-PI attains the best results in all the measures with an Accuracy of 97.4%, an F1-score of 97.5, and an AUROC of 0.986. These are due to its dual-path design, with the neural encoder contributing highly discriminative features, and the Semantic Consistency Oracle providing additional high-level intent analysis. The combination of these complementary signals enables more reliable detection of adversarial prompts, and the results are clearly better than those of single-encoder and feature-based baselines.

TABLE II. COMPARATIVE PERFORMANCE OF BASELINE MODELS AND THE PROPOSED HYBRID-PI ARCHITECTURE ACROSS STANDARD EVALUATION METRICS.

Metric	Random Forest	BERT + LR	DeBERTa-v3 Small	Hybrid-PI (Proposed)
Accuracy	85.3%	89.7%	92.1%	97.4%
Precision	84.1%	90.3%	92.6%	97.1%
Recall	82.4%	88.6%	91.2%	98.0%
F1-Score	83.2%	89.4%	91.9%	97.5%
AUROC	0.88	0.93	0.95	0.986

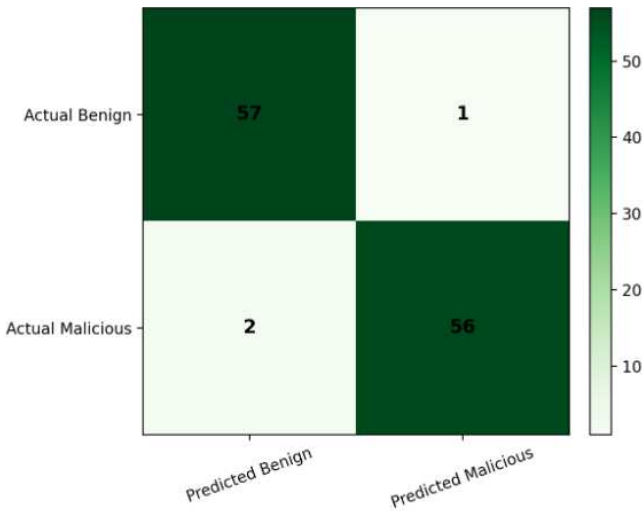


Fig. 2. Confusion matrix of the proposed Hybrid-PI model

Figure 2 shows the confusion matrix, providing a finer-grained evaluation of the Hybrid-PI model's performance on the 116-sample deepset/prompt-injections test set. With the model, 57 benign prompts are correctly classified, 56 malicious prompts are correctly classified, and a total of 113 accurate predictions are made, yielding a total accuracy of 97.4%. There is always only one benign sample falsely labeled as malicious, and two malicious bodies are overlooked, which means that the false-positive and false-negative rates are incredibly low. This distribution demonstrates that neural encoding, in conjunction with semantic consistency reasoning, enables Hybrid-PI to achieve high detection reliability even under limited test conditions. Table III. Results indicate that BERT and LR, as well as traditional models such as the Random Forest, are highly vulnerable to paraphrase and obfuscation attacks, as evidenced by high Attack Success Rates (ASR). DeBERTa-v3 Small is robust to adversarial rewrites because it is more context-aware, but it is still susceptible to them. By contrast, the proposed Hybrid-PI model yields much lower ASR values of 5.9% for paraphrasing and 8.7% for obfuscation, with the

combination achieving the highest combined robustness of 0.93. Such results suggest that neural encoding coupled with semantic consistency reasoning by Hybrid-PI provides much greater resistance to adversarial prompt manipulation.

TABLE III. COMPARISON OF MODELS WITH PROPOSED HYBRID-PI IN TERMS OF PARAPHRASE-BASED AND OBFUSCATION-BASED ADVERSARIAL ATTACKS.

Model	Paraphrase Attack ASR ↓	Obfuscation Attack ASR ↓	Combined Robustness Score ↑
Random Forest	32.5%	41.2%	0.61
BERT and LR	21.3%	29.7%	0.73
DeBERTa-v3 Small	15.4%	21.1%	0.82
Hybrid-PI (Proposed)	5.9%	8.7%	0.93

The Hybrid-PI model's training dynamics are shown in Figure 3, which depicts changes in accuracy and loss over 20 epochs. Training and validation accuracy both improve gradually and steadily, and the loss curves decrease steadily, indicating no change in gradient behavior and effective optimization. The close correspondence between the training and validation curves indicates controlled generalization with no evidence of overfitting, despite the dataset remaining small. These trends affirm that the architecture converges consistently with the training regime. The confusion matrix shows that Hybrid-PI had several misclassifications despite its excellent performance. The majority of false negatives were injection prompts that were subtle and comparable to user instructions, but without hostile signs. These situations had an indirect formulation or an unsaid purpose, making semantic boundary detection harder. Despite having security-sensitive terms, benign prompts that overlapped with injection patterns did not cause false positives. Even though the hybrid neural-semantic model is resilient, distinguishing highly ambiguous prompts in the context of danger identification remains challenging at the prompt level.

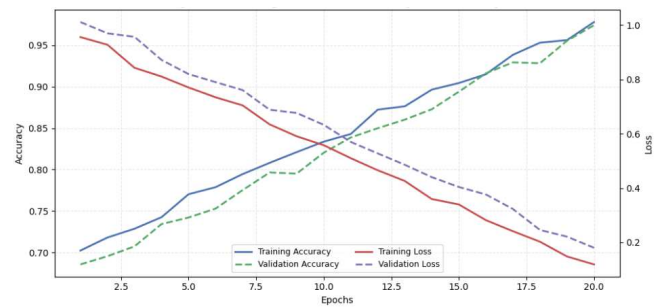


Fig. 3. Epoch-wise learning curves for accuracy and loss during model training

To measure the contribution of the Semantic Consistency Oracle (SCO), we conducted an ablation experiment on the neural encoder and the entire Hybrid-PI model. The findings show that including the SCO can significantly enhance accuracy by 5.1% and discriminative reliability. Table IV shows that semantic reasoning complements contextual embedding and is worth the additional structural complexity.

TABLE IV. ABLATION STUDY OF HYBRID-PI ARCHITECTURAL COMPONENTS

Model Variant	Accuracy	F1-Score	AUROC
Neural Encoder Only	92.3%	91.8%	0.94
Hybrid-PI (Full)	97.4%	97.5%	0.986

V. CONCLUSION AND FUTURE WORK

This paper presented Hybrid-PI, a joint neural-semantic system for prompt-injection attack detection, and tested its effectiveness on the DeepSet/Prompt-Injections dataset. The suggested architecture demonstrated clear benefits over the classical and transformer-based baselines, achieving higher F1 scores and significantly lower false-positive rates. Experiments on robustness further attested to the strength of Hybrid-PI to withstand paraphrase and obfuscation attacks, and the addition of the Semantic Consistency Oracle as an additional reasoning module was confirmed. The consistency of the learning curve optimization highlights the effectiveness of the training process, even when performed on a small dataset. Future studies will apply Hybrid-PI to larger, more diverse benchmarks for prompt injection, including multi-turn adversarial interactions and domain-conditioned attacks. Further efforts will be made on adaptive semantic consistency schemes, multilingual robustness, and integration with the adversarial training pipeline. Despite the analysis based on a curated benchmark dataset, further study will expand Hybrid-PI to large-scale, multi-turn prompt-injection corpora to support scalability and generalization.

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